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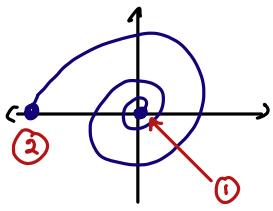
Lecture 28 4/1/26

More than two variables



write path as a parametric function $x = x(u)$ $y = y(u)$

$$L = \int ds = \int_{u_1}^{u_2} \sqrt{x'(u)^2 + y'(u)^2}\, du$$

$$S = \int_{u_1}^{u_2} f(x, y, x', y', u)\, du$$

$$X = x(u) + \alpha \xi(u)$$

$$Y = y(u) + \beta \eta(u)$$

Stationary point $\frac{\partial S}{\partial \alpha} = \frac{\partial S}{\partial \beta} = 0$

Derivation with x & y:

$$\frac{\partial S}{\partial \alpha} = 0 = \int_{u_1}^{u_2} \left(\xi \frac{\partial f}{\partial x} + \xi' \frac{\partial f}{\partial x'} \right) du = \int_{u_1}^{u_2} \left(\xi \frac{\partial f}{\partial x} - \xi \frac{d}{du} \frac{\partial f}{\partial x'} \right) du = \int_{u_1}^{u_2} \xi \left(\frac{\partial f}{\partial x} - \frac{d}{du} \frac{\partial f}{\partial x'} \right) du$$

$$\Rightarrow \frac{\partial f}{\partial x} - \frac{d}{du} \frac{\partial f}{\partial x'} = 0$$

$$\frac{\partial S}{\partial \beta} = 0 = \int_{u_1}^{u_2} \left(\eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'} \right) du = \int_{u_1}^{u_2} \left(\eta \frac{\partial f}{\partial y} - \eta \frac{d}{du} \frac{\partial f}{\partial y'} \right) du = \int_{u_1}^{u_2} \eta \left(\frac{\partial f}{\partial y} - \frac{d}{du} \frac{\partial f}{\partial y'} \right) du$$

$$\Rightarrow \frac{\partial f}{\partial y} - \frac{d}{du} \frac{\partial f}{\partial y'} = 0$$

$$L = \int_{u_1}^{u_2} \sqrt{x'(u)^2 + y'(u)^2}\, du$$

$$\frac{\partial f}{\partial x} = 0 \qquad \frac{\partial f}{\partial x'} = \frac{1}{2} \left[(x')^2 + (y')^2 \right]^{-1/2} 2x' = \frac{x'}{\sqrt{(x')^2 + (y')^2}}$$

$$\frac{\partial f}{\partial y} = 0 \qquad \frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{(x')^2 + (y')^2}}$$

$$\frac{\partial f}{\partial x} = \frac{d}{du}\frac{\partial f}{\partial x'} \qquad \frac{\partial f}{\partial y} = \frac{d}{du}\frac{\partial f}{\partial y'}$$

$$0 = \frac{d}{du}\frac{x}{\sqrt{(x')^2+(y')^2}} \qquad 0 = \frac{d}{du}\frac{y'}{\sqrt{(x')^2+(y')^2}}$$

$$C_1 = \frac{x}{\sqrt{(x')^2+(y')^2}} \qquad C_2 = \frac{y'}{\sqrt{(x')^2+(y')^2}}$$

$$\frac{C_1}{C_2} = \frac{x'}{y'} = m \qquad y' = mx'$$

$$\frac{dy}{\cancel{du}} = m\frac{dx}{\cancel{du}}$$

$$\boxed{y = mx + b}$$

In general, we will have many coordinates:

$$\mathcal{L}(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t)$$

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)\, dt \Rightarrow n \text{ different Euler-Lagrange equations}$$

$$\frac{\partial \mathcal{L}}{\partial q_n} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_n}$$ all of the same form.

Chapter 7: Lagrange's Equations

Consider a particle moving in 3D, subject to $\vec{F}(\vec{r})$, where $\vec{F}$ is conservative.

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\,\dot{\vec{r}}\cdot\dot{\vec{r}} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

$$U = U(\vec{r}) = U(x, y, z)$$

We can define a <u>Lagrangian</u> function:

$$\mathcal{L} = T - U$$

$\mathcal{L}$ determines the motion of the system in a very general way. You can think of $\mathcal{L}$ as a quantity that balances T & U.

Side note: <u>Very</u> roughly, you can think of $\mathcal{L}$ as coming from the following:

$\int p\,dx \rightarrow$ variations on this give different system configurations when varying x

$\int H\,dt \rightarrow$ variations on this give different system configurations when varying t

$$\underset{\text{variation}}{\delta}\left[\int p\,dx\right] = \underset{\text{variation}}{\delta}\left[H\,dt\right] \Rightarrow$$ the system's configuration in x & in t should match if $x(t)$ is the correct trajectory

$$S = \int p\,dx - \int H\,dt \quad \text{and} \quad \delta S = 0 \quad \text{(Action is at a stationary point)}$$

$$\left.\begin{aligned} p\,dx &= mv\,dx \\ &= mv\frac{dx}{dt}\,dt \\ &= mv^2\,dt \\ &= 2T\,dt \end{aligned}\right\}$$ ultimately becomes $2T\,dt$

ultimately becomes $(T+U)\,dt$

$$(2T) - (T+U) = T - U$$

$$S = \int \mathcal{L}\,dt = \int (T-U)\,dt \quad \Rightarrow \quad \mathcal{L} = T - U$$

[end rough justification of $\mathcal{L}$]

Let's examine:

- $T(\dot{x}, \dot{y}, \dot{z})$ & $U(x,y,z) \Rightarrow \mathcal{L}(x, y, z, \dot{x}, \dot{y}, \dot{z})$

- $$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial}{\partial x}(T-U) = -\frac{\partial U}{\partial x} = F_x$$
$$-\frac{\partial U}{\partial y} = F_y$$
$$-\frac{\partial U}{\partial z} = F_z$$

- $$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial}{\partial \dot{x}}(T-U) = m\dot{x} = p_x$$
$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = p_y$$
$$\frac{\partial \mathcal{L}}{\partial \dot{z}} = p_z$$

- $$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{d}{dt}(m\dot{x}) = m\ddot{x} = ma_x$$
$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}} = ma_y$$
$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{z}} = ma_z$$

Newton's 2nd law:

$$\vec{F} = m\vec{a} : \quad F_x = m\ddot{x} \iff \frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} \quad \Big\} \text{ Lagrange eqns!}$$

From Chapter 6:

$$S = \int_{x_A}^{x_B} f(y_1, y_2, y_3, y_1', y_2', y_3', x)\,dx \qquad\qquad S = \int_{t_1}^{t_2} \mathcal{L}(x, y, z, \dot{x}, \dot{y}, \dot{z}, t)\,dt$$

is stationary when $\dfrac{\partial f}{\partial y_i} = \dfrac{d}{dt}\dfrac{\partial f}{\partial y_i'}$ $\Longrightarrow$ $\dfrac{\partial \mathcal{L}}{\partial x} = \dfrac{d}{dt}\dfrac{\partial \mathcal{L}}{\partial \dot{x}}$ (y, z)

<u>Hamilton's Principle</u>: The path a particle follows between points 1 & 2 in the time interval t_1, t_2 is such that the <u>action</u> is stationary:

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)\,dt$$

The action

Hamilton's principle lets us use $\mathcal{L}$ equations hold in any coordinate system.

We can use Cartesian (x, y, z); Cylindrical (ρ, ϕ, z); Spherical (r, θ, ϕ), etc.

Consider a generalized coordinate system q_1, q_2, q_3:

$$q_i = q_i(\vec{r}) \quad , \quad i = 1, 2, 3$$

$$\vec{r} = \vec{r}(q_1, q_2, q_3)$$

[Each $\vec{r}$ gives a unique q_1, q_2, q_3 and each (q_1, q_2, q_3) gives a unique $\vec{r}$]

$$\mathcal{L} = \tfrac{1}{2} m \dot{\vec{r}}^{\,2} - U(\vec{r}) = \mathcal{L}(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3)$$

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3)\,dt$$

S = stationary $\Rightarrow$

$$\frac{\partial \mathcal{L}}{\partial q_1} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_1}$$

$$\frac{\partial \mathcal{L}}{\partial q_2} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_2}$$

$$\frac{\partial \mathcal{L}}{\partial q_3} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_3}$$

$$\Longrightarrow \quad \underbrace{\frac{\partial \mathcal{L}}{\partial q_i}}_{\text{generalized force}} = \frac{d}{dt}\underbrace{\frac{\partial \mathcal{L}}{\partial \dot{q}_i}}_{\text{generalized momentum}}$$